



CRYSTAL CALIBRATION SALES AND SERVICE CO., LTD.

45/48 Soi Salathammasop31, Salathammasop Rd.,

Salathammasop, Thawewatthana, Bangkok 10170 Thailand

Tel : 0-2408-8474-5 Fax : 0-2408-8477 Email : info@crystalcal.com www.crystalcal.com



CERTIFICATE OF CALIBRATION

Certificate No. : 25-1390-078

Issue Date : 31 October 2025

Work Order No. : 25/1390

Customer Name : กลุ่มงานเทคนิคการแพทย์ โรงพยาบาลพระจอมเกล้า จังหวัดเพชรบุรี
53 ถนนไฟ ต.คลองกระแซง
อ.เมือง จ.เพชรบุรี 76000

Date of Received : 27 October 2025

Date of Calibration : 27 October 2025

Instrument Details : Description : Temperature Controlled Enclosures [Refrigerator]
Manufacturer : SANDENINTERCOOL
Model : SPA-0303D41A
Serial No. : SPA0303-190602731
ID No. : 4110-007-0002-1/63
Resolution : N/A
Location : Special Lab

Calibration Method : This instrument was calibrated by insert standard thermometer into the chamber according to calibration procedure no. CWI-T-10 follow up to TLAS G-20-1/02-08 (E) : Guidelines for Calibration and Checks of Temperature Controlled Enclosures.

Environmental Conditions :

Temperature : Area Monitoring between 15°C to 40°C

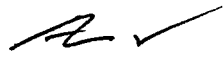
Humidity : Area Monitoring between 30%RH to 85%RH

Line Voltage : Area Monitoring 220 VAC $\pm$ 10%

Traceability of Measurement :

This certificate of calibration documents the traceability to national standard which realize the unit of measurement according to the International system of Units (SI) and The temperature scale in use at this laboratory is The International Temperature scale of 1990.

Calibrated by : Mr. Nattaphon Boonrod
Calibration Engineer

Approved by : 
(Mr. Anuwat Yaklermjit)
Laboratory Manager

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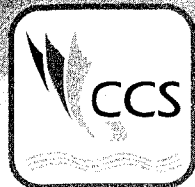
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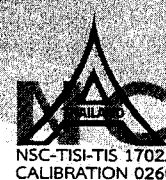


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Details of Calibration

1. Reference Standards Instrument

Instrument	Model	Serial No./Ins No.	Certificate No.	Due Date
Data Acquisition unit	34972A	MY49024826	25-0861-020	06 July 2026
Sensor type	RTD	CH # 301-309	25-0861-020	06 July 2026

2. Certificate traceble

: This certificate traceable to The International System of Unit refer to
Crystal Calibration Sales and Service Co., Ltd. , NSC-ONSC Calibration No. 0260

3. Condition of item

: Used

4. Calibration site

: On - Site

5. Result of Calibration

: Without adjustment

6. Evaluate Condition

: Time Constant : - Hour 33 Minute At cal. point 4 °C
Air vent : Off
Fan speed status : Fixed Fan Speed

7. Calibration note

: The results reported in this certificate refer to the condition of instrument on
the process into the steady state of chamber

8. Sensors Installation Diagram

: When ; Sensor installation location in Chamber @ Working Space
A = Distance between sensor and wall of chamber is 10 cm

9. Dimensions of chamber

: W = 0.6 m ; D = 0.55 m ; H = 1 m

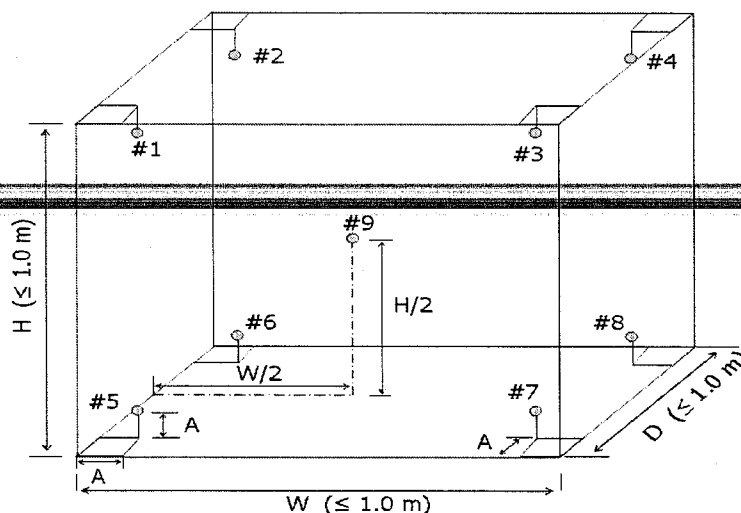


Diagram of Chamber



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Result of Temperature Distribution and Performance Check

Table 1 : Reporting of Temperature Distribution

Calibration point (°C)	Average Measured Temperature (°C) @ Sensor No. (Sensor No.9 is REF)									Uncertainty ± (°C)
	#1	#2	#3	#4	#5	#6	#7	#8	#9	
4	6.16	5.95	5.80	5.60	5.78	5.82	6.26	5.91	5.26	2.7

Table 2 : Reporting of Performance check

Calibration Point (°C)	Indicator Set Point (°C)	Indicator Reading (°C)			Stability ± (°C)	Uniformity (°C)	Overall variation (°C)
		MAX	MIN	Average			
4	As mark	N/A	N/A	N/A	2.22	1.89	5.93

Note

Calibrate items in good condition and this report customer request and accepted in certificate

The reference sensor is preferably located at the geometric center of chamber

The measured temperature data readout by software "Benchlink Datalogger 3"

The quoted uncertainty include "Stability" and "Loading effect (20% of Temp Uniformity)"

Stability - one-half of the greatest maximum difference of measured temperatures at any one sensor.

Uniformity - the maximum difference of measured temperatures at any sensors and the measured temperature at the reference location which are observed at the same time or at as close an observation time as possible to determine the temperature pattern or homogeneity within the chamber under steady state conditions.

Overall Variation - The difference of the maximum and minimum measured temperatures throughout observation time.

Indicating Temperature - the average reading of indicating device that forms the integral part of the enclosure.

This result of calibration was found accurate as shown on date and place of calibration only.

The reported uncertainty of measurement was based on a standard uncertainty multiplied by a coverage factor $k=2$ providing a level of confidence of approximately 95%.



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Statements of conformity report

Certificate No. : 25-1390-078

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Detail of Equipment

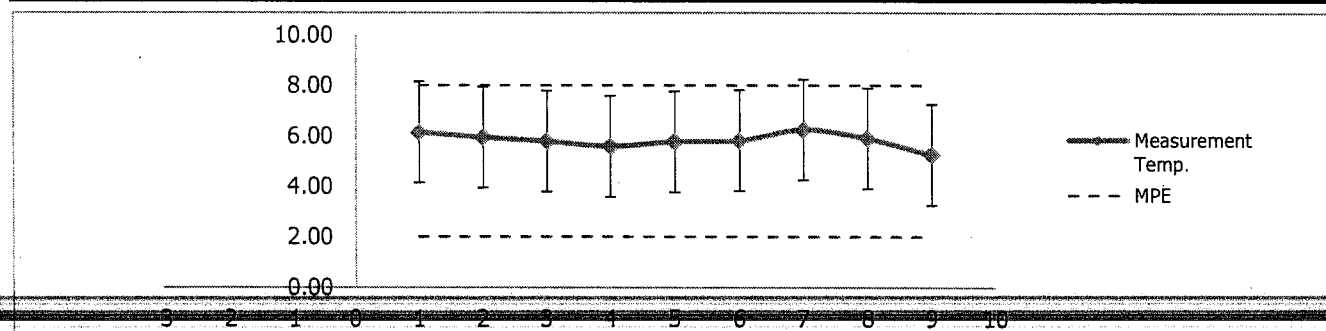
Description : Temperature Controlled Enclosures [Refrigerator]
Manufacturer : SANDENINTERCOOL ID No. : 4110-007-0002-1/63
Model : SPA-0303D41A Serial No. : SPA0303-190602731

Result of Calibration

This result of calibration : Without adjustment

Calibration point	4	°C	Uncertainty	Uncertainty MPE	Evaluate resulte
Customer MPE	2-8	°C	2.7	°C	3.0
					Passed

Evaluate upper limit [UL]				Evaluate lower limit [LL]			
Position	MEAN STD	MPE	Lab. Decide	Position	MEAN STD	MPE	Lab. Decide
#1	6.16	8.0	Pass	#1	6.16	2.0	Pass
#2	5.95	8.0	Pass	#2	5.95	2.0	Pass
#3	5.80	8.0	Pass	#3	5.80	2.0	Pass
#4	5.60	8.0	Pass	#4	5.60	2.0	Pass
#5	5.78	8.0	Pass	#5	5.78	2.0	Pass
#6	5.82	8.0	Pass	#6	5.82	2.0	Pass
#7	6.26	8.0	Pass	#7	6.26	2.0	Pass
#8	5.91	8.0	Pass	#8	5.91	2.0	Pass
#9	5.26	8.0	Pass	#9	5.26	2.0	Pass



This statement judgment and decision by Laboratory guideline only

The conformity certificate documents validity following ISO/IEC Guide 98-4 : Role of measurement uncertainty in conformity assessment based on statement with guard band refer to AS 3864 consider expanded measurement uncertainty (k=95%)

The tolerance and decision rules ;

Decision rules : Laboratory decide temperature measurement in chamber is ;

MPE of Customer or Lab decide = Measurement temperature of any points ; [$\leq$ MPE is pass , $>$ MPE is Fail]

Statements of conformity decision by :

(Mr. Anuwat Yaklermjit)

Laboratory Manager

-- END --